

■# Business Report: HR Attrition Risk Prediction Model

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Audience: MNC Business Heads and HR Leadership

Project: HR Attrition Risk Prediction and Dashboard

Updated after latest project changes

1. Executive Summary

This project helps HR and business leaders identify employees who may be at higher risk of leaving the company. It uses employee data such as department, job role, salary, overtime, satisfaction, work-life balance, years at company, years with manager, and promotion-related information.

The project has now been organized in a cleaner way. The dashboard is inside app/, the data is inside data/, generated reports are inside reports/generated/, and the optional graph/Neo4j layer is separated from the core ML scoring pipeline. This makes it easier for a business or technical team to understand, maintain, and audit.

Current business snapshot:

Business Metric	Current Value
Active employees scored	1,489
High-risk employees above 30% threshold	1,022
Average 12-month leave risk	61.21%
Largest department	Research & Development
Main business output	Risk probability for 1M, 3M, 6M, and 12M

2. What Problem This Solves

Employee attrition is costly for an MNC because it can cause:

- Project delivery delays
- Client knowledge loss
- Higher hiring and replacement cost
- More workload on remaining employees
- More onboarding and training effort
- Risk in critical teams and leadership pipelines

This model works like an early warning system. It helps the company notice risk before resignation happens.

3. What the Model Shows

The model does not only say whether an employee may leave or not. It gives time-based risk.

Risk Period	Business Meaning
1 month	Immediate attention required
3 months	Short-term retention action
6 months	Medium-term HR planning
12 months	Annual workforce planning

This is helpful because HR teams can prioritize urgent cases separately from long-term cases.

4. Current Business Charts

Image: images/business_risk_charts.svg

From the current scored data:

- Average 1-month leave risk is 39.94%.
- Average 3-month leave risk is 47.43%.
- Average 6-month leave risk is 53.86%.
- Average 12-month leave risk is 61.21%.
- 1,022 out of 1,489 employees are above the 30% high-risk threshold.

5. Department-Level View

Department	Employees	Average 12-Month Risk	High-Risk Employees
Human Resources	63	63.04%	44
Research & Development	975	63.57%	702
Sales	451	55.87%	276

Business interpretation:

- Research & Development has the highest number of high-risk employees because it has the largest employee count.
- Human Resources and Research & Development both show high average 12-month risk, so they should be reviewed closely.
- Sales should also be watched because attrition in Sales can affect clients and revenue continuity.

6. How Business Teams Can Use It

Image: images/business_usage_flow.svg

Simple business usage:

1. Open the HR dashboard.
2. View total active employees and high-risk count.
3. Filter by department or high-risk status.
4. Sort employees by risk score.
5. Open employee detail drawer.
6. Check risk drivers, risk curve, and graph exposure context when available.
7. Run what-if actions like salary hike, overtime change, manager change, or promotion.
8. Decide whether HR should take action.
9. Re-score after action to see updated risk.

7. Business Meaning of Risk Drivers

The dashboard groups risk drivers into five themes.

Theme	Meaning for Business
Identity	Employee background fields like age, education, gender, travel
Environment	Department, job role, overtime, distance, job level
Compensation	Salary, salary hike, stock option, pay rates
Sentiment	Satisfaction, work-life balance, involvement, performance
Tenure	Years at company, promotion gap, manager tenure, experience

This helps HR understand the type of issue behind the risk.

Example:

- If compensation is high, salary or benefits may need review.
- If environment is high, manager, role, overtime, or team situation may need review.
- If sentiment is high, employee engagement may need action.
- If tenure is high, promotion delay or career growth may be the issue.

8. What-If Planning

The dashboard supports what-if checks. This means HR can test a possible action before applying it.

Examples:

What-If Action	Business Question
Salary hike	Will better compensation reduce risk?
Overtime change	Does reducing overtime lower risk?
Manager tenure change	Could manager/team change improve risk?
Promotion/job role change	Would career growth reduce attrition risk?

This is useful because it gives business leaders a data-supported way to compare retention actions.

9. Network Exposure Context

The project also includes an optional graph layer for understanding whether an employee is close to recently exited peers. This is not used as a stored model feature in the main production scoring table because the validation work did not prove strong contagion lift. Instead, it is used as dashboard context.

Business meaning:

Graph Signal	Business Use
Same manager	Check team-level retention pressure
Same department and role	Check peer group attrition pressure
Same tenure cohort	Check career-stage attrition pressure
Recent connected exits	Understand whether risk may be socially or operationally amplified

If Neo4j is offline, the dashboard keeps working with a mock fallback. This keeps the business demo stable while still showing where graph intelligence can fit later.

10. How This Can Help an MNC

The model can help in these areas:

Area	Benefit
Retention planning	Focus on employees with higher risk
HR budgeting	Use salary/engagement budget more carefully
Workforce planning	Understand future attrition pressure by department
Manager review	Identify teams where environment factors may be high
Employee engagement	Find groups where satisfaction-related risk is visible
Hiring planning	Prepare replacement hiring where risk is very high

11. Business Governance and Care

This model should support human decision-making. It should not become the final decision maker.

Recommended rules:

- Use the model for support and retention planning only.
- Do not punish employees based on model risk.
- Keep employee-level risk restricted to approved HR/business leaders.
- Review fairness across departments, gender, age bands, and job roles.
- Track whether HR actions actually reduce attrition over time.
- Re-train and monitor the model regularly.

12. Suggested MNC Rollout Plan

Phase	Action
Phase 1	Run pilot with HR analytics team
Phase 2	Validate high-risk employees with HR business partners
Phase 3	Add security, audit logs, and access control
Phase 4	Connect to real HRMS source system
Phase 5	Schedule monthly or weekly scoring
Phase 6	Track intervention success and business impact

13. Final Business Conclusion

This project can help an MNC reduce attrition risk by giving early signals to HR and business leaders. It shows who may be at risk, when the risk may happen, and what kind of factor may be contributing.

In simple words, this model helps the company act before employee attrition becomes a bigger business problem.